Hill and Griffith Company
Safety Data Sheet

May be used to comply with OSHA's Hazard Communication Standard, 29 CFR 1910.1200. Standard must be consulted for specific requirements. This information should be given to all individuals who may come in contact with this material.

N.A. = NOT APPLICABLE N.E. = NOT ESTABLISHED

1.0 CHEMICAL PRODUCT AND COMPANY IDENTIFICATION

PRODUCT NAME (s): Mold Seal
Product use: Caulking Compound
Description: Mold Sealant

SUPPLIER: The Hill & Griffith Co.
1085 Summer Street
Cincinnati, Ohio 45204
(513) 921-1075 Fax:(513) 921-9180

EMERGENCY TELEPHONE: (800) 424-9300
Call Chemtrec

2.0 HAZARD IDENTIFICATION

Emergency Overview

Appearance	Smooth Paste
Color	Gray
Odor	odorless

This material is considered hazardous by the OSHA Hazard Communication Standard (29 CFR 1910.1200)

GHS Classification

Human Health

Acute Toxicity: Oral – Category 4
Serious Eye Damage / Irritation – Category 2B
Specific Target Organ Toxicity (Single Exposure) [central nervous system (CNS) and kidneys] – Category 2
Specific Target Organ Toxicity (Repeated Exposure) [central nervous system (CNS) kidneys and liver] – Category 2

GHS Label element

Hazard Pictograms

Signal word: **Danger**

Hazard Statements
H302 Harmful if swallowed
H320 Causes eye irritation

	H371 May cause damage to organs, (central nervous system (CNS), kidneys) H373 May cause damage to organs through prolonged or repeated exposure. (central nervous system (CNS), kidneys, liver)
Precautionary statements	Do not breathe dust/fume/gas/mist/vapors/spray. In case of inadequate ventilation wear respiratory protection. Wear eye or face protection. Recommended: Wear suitable safety spectacles. Do not eat, drink or smoke when using this product. Wash hands thoroughly after handling. IF exposed or you feel unwell: Call a Poison Center or Physician. IF SWALLOWED: Call a Poison Center or Physician if you feel unwell. Rinse Mouth. IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing. If eye irritation persists: Get medical attention. Store locked up. Dispose of contents and containers in accordance with local, regional, national and international regulations.

Potential Health Effects

Principle Routes of Exposure:	Inhalation, Skin contact, Eye contact, Ingestion
Aggravated Medical Condition:	None known.

3.0 COMPOSITION / INFORMATION ON INGREDIENTS

CHEMICAL NAME	CAS – No	Weight%
Ethylene glycol	107-21-1	10 - 15
Bentonite	1302-78-9	10 - 15

4.0 FIRST-AID MEASURES

INHALATION: Move to fresh air. If not breathing, give artificial respiration. If breathing is difficult, give oxygen. If symptoms persist, obtain medical attention..

SKIN CONTACT/ABSORPTION: Wash skin thoroughly with soap and water. Remove contaminated clothing. If irritation develops, seek medical attention.

EYE CONTACT: Flush with large amounts of clear flowing water for at least 15 minutes. As a precaution remove any contact lenses, if worn. Hold eyelids apart to ensure rinsing of the entire surface of the eye and lids with water. If irritation persists, seek medical attention.

INGESTION: Wash out mouth with water. Remove dentures if anyDo NOT induce vomiting. As a precaution get medical advice. If spontaneous vomiting occurs, place the victim's head below knee level. Never give anything by mouth to an unconscious person.

PROTECTION OF FIRST_AIDERS: No action shall be taken involving any personal risk or without suitable training.

NOTE TO PHYSICIAN: Treat symptomatically. Contact poison treatment specialists immediately if large quantities have been ingested or inhaled.

5.0 FIRE FIGHTING MEASURES

FLASHPOINT: (Cleveland Open Cup) Not combustible

FLAMMABLE LIMITS **LEL:** NA **UEL:** NA

AUTOIGNITION TEMPERATURE: NE

EXTINGUISHING MEDIA: Carbon dioxide, foam, dry chemical, water fog

UNUSUAL FIRE AND EXPLOSION HAZARDS: In a fire or if heated, a pressure increase will occur and the container may burst.

FIRE FIGHTING EQUIPMENT: Wear self-contained breathing apparatus with a full face-piece operated in the positive pressure demand mode when fighting fires.

6.0 ACCIDENTAL RELEASE MEASURES

Use proper respiratory and personal protective equipment. MSHA/NIOSH or OSHA/NIOSH approved respirator recommended. Spilled material may cause slippery conditions when wet. Care should be exercised when walking on spills on floors or concrete pads. Vacuum, pump or scoop spilled material into containers for reclaiming or disposal. Do not discharge into drains, watercourses or onto ground.

7.0 HANDLING AND STORAGE

HANDLING & STORAGE: Put on appropriate protective equipment. Do not get in eyes or on skin or clothing. Do not ingest. Eating, drinking and smoking should be prohibited in areas where this material is stored and processed. Workers should wash hands and face before eating, drinking or smoking. Remove contaminated clothing and protective equipment before entering eating areas. Store in a cool and well ventilated place. Keep away from incompatible materials. Keep container closed and sealed until ready for use. Do not store in unlabeled containers.

8.0 EXPOSURE CONTROLS AND PERSONAL PROTECTION

INHALATION PROTECTION: Not normally required but if respirator is required, use of a MSHA/NIOSH or OSHA/NIOSH approved respirator is recommended.

EYE PROTECTION: Use safety glasses with side shields or goggles when transferring material to prevent material from splashing into the eye.

SKIN PROTECTION: Rubber gloves are recommended for prolonged exposure.

ENGINEERING CONTROLS: Local exhaust at point of generation. Local or general exhaust required when using material at elevated temperatures, which accelerates generation of vapors.

EXPOSURE GUIDELINES:

Ingredient	CAS Number	Exposure Limits
Ethylene glycol	107-21-1	100 mg/m^3 (ACGIH TLV) Form: aerosol
Bentonite	1302-78-9	1 mg/m^3 (ACGIH TLV-TWA)

9.0 CHEMICAL AND PHYSICAL INFORMATION

Appearance:	Gray Paste
Odor:	None

Water Solubility:	Miscible
Melting Point(°F):	32
Boiling range:	212°F
Flammability:	Not determined
Auto-ignition temperature:	Not determined
Decomposition temperature:	Not determined
Auto-igniting:	Product is not self-igniting
Danger of explosion:	Product does not present an explosion hazard
Explosion limits:	Upper: Not determined
	Lower: Not determined
Relative Density:	Not Determined
Specific Gravity (water=1):	1.1
Vapor Pressure (mmHg):	760
pH:	NA
Vapor Density (air=1):	NE
Viscosity:	
Dynamic:	Not Determined
Kinematic:	Not Determined
Other Information:	No additional information

10.0 STABILITY AND REACTIVITY

STABILITY: Stable under normal temperature and pressure.

CONDITIONS TO AVOID: Avoid heat, sparks, flame, and strong oxidizing agents

MATERIALS TO AVOID: Strong oxidizers, strong acids

HAZARDOUS DECOMPOSITION: Thermal decomposition is highly dependent on the combustion conditions. A complex mixture of airborne solid, liquid, particulate, and gases will evolve when this material undergoes pyrolysis or combustion. Oxides of carbon, and other unidentified organic compounds may be formed upon combustion.

HAZARDOUS POLYMERIZATION: Will not occur.

11.0 TOXICOLOGICAL INFORMATION

Skin corrosion/irritation: May cause mechanical skin irritation.
Serious eye damage/eye irritation: Causes eye irritation
Respiratory or skin sensitization
Respiratory sensitization Not classified
Skin Sensitization Not classified
Germ cell mutagenicity Not classified
Carcinogenicity: While product contains a small amount of crystalline silica it is not in a respirable form.
US. OSHA Specifically Regulated Substances (29 CFR 1910.1001-1050): Not listed
Reproductive toxicity: Not classified
Teratogenicity: Positive tests on both male, female mice and male, female rats by oral and negative by dermal
Specific target organ toxicity – Single exposure Category 2 – Central nervous system and kidneys
- Repeated exposure Category 2 Central nervous system , kidneys and liver.

Acute Toxicity

Eyes: Cause eye irritation, watering and redness

Skin: May cause mechanical skin irritation. Contact with the skin is not expected to cause an allergic reaction. Not expected to be harmful to internal organs if absorbed through the skin. Wash thoroughly after handling.

Inhalation: Long term exposure to crystalline silica can cause lung injury (silicosis). IARC and NTP have determined that crystalline silica inhaled from occupational sources can cause cancer in humans. Risk of injury is dependent on the duration and level of exposure.

Ingestion: Harmful if swallowed. May be irritating to mouth, throat and stomach.

Acute Toxicity Data

EYE IRRITATION:

SKIN IRRITATION:

DERMAL LD50: Ethylene glycol: 7712 mg/kg Male and female rat.
Bentonite: No data available

ORAL LD50: Ethylene glycol: LD50 >3500 mg/kg Male and female mice.
Bentonite: > 5000 mg/kg Rat, > 2000 mg/kg Rat

INHALATION LC50: Bentonite: >5.27 mg/L Rat

Potential chronic health effects:

Ethylene glycol: OECD 408 Repeated Dose 90 day Oral Toxicity Study in Rodents: 150mg/kg/d Sub-chronic NOAEL oral, Rat male, female.
200 mg/kg/d Chronic NOAEL Oral, Rat male, female
1500 mg/kg/d Chronic NAOEL Oral, Mouse, male, female

OECD 407 Repeated Dose 28 day Oral Toxicity Study in Rodents: 200 mg/kg/d Sub-acute NOAEL Oral Rat, male, female

OECD 410 Repeated Dose Dermal Toxicity 21/28 day Study: > 4000 mg/kg Sub-acute NOEL Dermal, Dog, Male

OECD 410 Repeated Dose Dermal Toxicity 21/28 day Study: >2000 mg/kg Sub-acute NOAEL Dermal, Dog, male, female

Acute Toxicity Estimates: Route: Oral ATE value: 505.1 mg/kg

OTHER TOXICITY DATA: The animal toxicity data for ethylene glycol (EG) and diethylene glycol (DEG) does not reflect the human toxicity of these substances. EG and DEG can be harmful or fatal if swallowed. The lower bound of human toxicity for these substances is estimated to bo 200 mg/kg.

12.0 ECOLOGICAL INFORMATION

TOXICITY:

Ethylene Glycol: OECD 202 Daphnis Acute Immobilization test, EC50: >100 mg/l 48 hour Daphnia

Acute ErC50 Growth rate, 6500 to 13,000 mg/l, 96 hour, Algae

Acute LC50, 72,860 mg/l, 96 hour static, fish

ISO 8192, Chronic, EC20 >1995 mg/l, 30 minutes, Bacteria

Chronic NOEC 8590 mg/l, 7 days static, Daphnia

Chronic NOEC 15380 mg/l, 7 days static, fish

Bentonite: EC50 >100 mg/l, 72 hours, freshwater algae

TLM96 10000 ppm, Oncorhynchus mykiss, fish

LC50 16000-19000 mg/l, 96 hours, Oncorhynchus mykiss, fish

LC50 2800-3200 mg/l, 24 hours, Black bass, blue gill, and sunfish

EC50 81.6 mg/l, 96 hours, Metacarcinus magister, invertibrates

EC50 24.8 mg/l, 96 hour, Pandalus danae, invertivrates

EC50 >100 mg/l, 96 hours, Daphnia magne, invertibrates

Persistence and degradability

Ethylene Glycol: OECD 301A Ready Biodegradability - Die-Away Test: 10 days, 90-100% -- Readily biodegradable

Bentonite: The methods for determining biodegradability are not applicable to inorganic substances.

Bioaccumulative potential

Ethylene Glycol: $LogP_{ow}$ + -1.36 Potential is low

Bentonite: No information available.

Mobility in Soil

Ethylene Glycol: Soil/water partition coefficient, K_{oc} = 0 – 1, no known significant effects or critical hazards.

Bentonite: no information available.

ENVIRONMENTAL FATE:

This material is not expected to be readily biodegradable. See Section 6 for accidental release measures.

13.0 DISPOSAL INFORMATION

Waste generated during application, demolition, breakage, or spillage may be hazardous waste as defined by RCRA (40 CFR Part 261). Place waste and spillage in closed containers. Dispose of in approved landfill in accordance with federal, state, and local regulations. Do not pollute. Conserve resources.

14.0 TRANSPORTATION INFORMATION

HAZARD CLASS:

IDENTIFICATION NUMBER: None

Shipping Name: Any Packaging Caulking Compound N.O.I.

15.0 REGULATORY INFORMATION

CERCLA SECTIONS 102A / 103 HAZARDOUS SUBSTANCES (40 CFR PART 302.4): listed

SARA TITLE III SECTION 302 EXTREMELY HAZARDOUS SUBSTANCES (40 CFR PART 344): Not Regulated

SARA TITLE III SECTION 311 / 312 HAZARDOUS CATEGORIZATION (40 CFR 370): Immediate and delayed health hazard

SARA TITLE III SECTION 311/312 MSDS Distribution – Chemical Inventory – hazard identification: None

SARA TITLE III SECTION 313 (40 CFR PART 372): Form R reporting Req. Ethylene Glycol – Concentration 10-15%

U.S. INVENTORY (TSCA): All ingredients listed on inventory or exempted.

OSHA HAZARD COMMUNICATION STANDARD: Under normal conditions this material is known to be hazardous.

16.0 OTHER INFORMATION

Label Requirements: NOT EXPECTED TO PRODUCE SIGNIFICANT ADVERSE HEALTH EFFECTS WHEN THE RECOMMENDED INSTRUCTIONS FOR USE ARE FOLLOWED.

HMIS RATINGS: Health: 1 Flammability: 0 Reactivity: 0

This Material Safety Data Sheet conforms to the requirements of ANSI Z400.1. As of the date of preparation of this document, the forgoing information is believed to be accurate and is provided in good faith to comply with applicable federal and state law(s). However, no warranty or representation with respect to such information is intended or given.

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Prepared by: Duane G. Wahl Title: Quality / Lab Services Manager

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